

How do we invert a linear map given by a matrix $A \in F^{n \times n}$? The map takes $x \in F^n$ as the input and produces $Ax = y$ as the output. So, to invert the map we need to find the "way back" - from y to x . For a given y , $Ax = y$ is a system of linear equations in the unknowns $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

So to invert a linear map, we need to learn how to solve linear systems.

Let's consider an arbitrary linear system with m equations and n unknowns $x_1, \dots, x_n$:

$$(LS) \quad \sum_{j=1}^n a_{ij} x_j = b_i \quad (i=1, \dots, m)$$

where $a_{ij}, b_i \in F$ are given coefficients of the system.

In the matrix form the system is written as

$$Ax = b$$

where $A = (a_{ij})_{\substack{i=1, \dots, m \\ j=1, \dots, n}} \in F^{m \times n}$, $b \in (b_i)_{i=1, \dots, m} \in F^m$

and $x = (x_j)_{j=1, \dots, n} \in F^n$ is unknown.

Bsp

$$\begin{cases} x_1 + 2x_2 = 4 & (1) \\ 3x_1 + 4x_2 = -1 & (2) \end{cases} \quad \begin{matrix} \leftarrow 3 \times \\ \leftarrow \end{matrix}$$

system over $\mathbb{Q}$

$$\begin{cases} x_1 + 2x_2 = 4 & (1') \\ \boxed{-2x_2 = -13} & (2') \end{cases}$$

$$\begin{array}{l} x_1 + 2x_2 = 4 \Rightarrow -3x_1 + 6x_2 = -12 \\ 3x_1 + 4x_2 = -1 \Rightarrow 3x_1 + 4x_2 = -1 \end{array} \Rightarrow -2x_2 = -13$$

If (1') and (2') are true, then (1) is also true,
because (1) is just the same as (1').

What equation do we get by adding $3 \times (1')$ to (2')?

$$\underbrace{3(x_1 + 2x_2)}_{\text{from (1')}} - \underbrace{2x_2}_{\text{from (2')}} =$$

$$\underbrace{3 \cdot 4}_{\text{from (1')}} - \underbrace{13}_{\text{from (2')}} =$$

$$3x_1 + 4x_2 = -1, \text{ which is equation (2)}$$

$$\left\{ \begin{array}{l} x_1 + 2x_2 = 4 \\ -2x_2 = -13 \end{array} \right. \quad \begin{array}{l} (1') \\ (2') \end{array} \quad \left. \begin{array}{l} \uparrow \\ \downarrow \end{array} \right\} +1.$$

$$\left\{ \begin{array}{l} x_1 = -9 \\ -2x_2 = -13 \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} x_1 = -9 \\ x_2 = \frac{13}{2} \end{array} \right.$$

Observations: it's nice to use transformation, of equations as above.

If a variable x_k occurs in exactly one equation, we are happy: it does not interfere with any other equations and we get its value from the equation, in which it occurs.

Def. We say that a variable x_k occurs in the equation $\sum_{j=1}^n a_{ij} x_j = b_i$ if $a_{ik} \neq 0$.

Def. We introduce the following operations on the equations:

- changing the order of two equations:
 $(s) \leftrightarrow (t)$ (swapping s -th and t -th equations)

- multiplying an equation with a non-zero number: $(s) := \gamma \cdot (s)$ for $\gamma \in F \setminus \{0\}$.
- Adding a multiple of one equation to another one:

$$(s) := (s) + \gamma \cdot (t)$$

to the s -th equation we add γ times t -th equation ($s \neq t$).

We call the above transformations of linear systems.

Proposition 30 Application of elementary transformations to a linear system does not change the solution set of a linear system. (That is, by carrying out elementary transformations one

arrives at an equivalent system.)

Let's develop a method that solves (LS).

We introduce J , the set of indices j of the variables that are contained in exactly one equation and the set I of the indices i of the equations that contain one variable.

We have $|I| = |J|$.

We want to make this property an invariant and we want to make I and J larger in each iteration.

$$\begin{cases} x_1 + 2x_3 - x_4 = 5 \\ x_2 - x_3 - x_4 = 4 \\ x_3 + x_4 = 10 \end{cases}$$

$$J = \{1, 2\}$$

$$I = \{1, 2\}$$

Algorithm. (Gaussian elimination)

$$I := \emptyset$$

$$J := \emptyset$$

while there exists a_{ij} with $a_{ij} \neq 0$ and $i \notin I, j \notin J$,
eliminate x_j from all the equations but the
 i -th one, add i to I and j to J .
(eliminate using elementary transformations).

Example

$$\begin{cases} x_1 + x_2 + x_3 = 5 & (1) \\ x_1 - x_2 + x_3 = 1 & (2) \\ x_1 + 2x_2 - x_3 = 0 & (3) \end{cases}$$

$$I = \emptyset$$

$$J = \emptyset$$

$$\begin{cases} x_1 + x_2 + x_3 = 5 & (1) \\ -2x_2 = -4 & (2) \\ x_2 - 2x_3 = -5 & (3) \end{cases}$$

$$I = \{1, 3\}$$

$$J = \{1, 3\}$$

$$\left\{ \begin{array}{l} x_1 \rightarrow 3x_3 = +10 \quad (1) \\ -7x_3 = -14 \quad (2) \\ x_2 - 2x_3 = -5 \quad (3) \end{array} \right.$$

$$I = \{1, 3\}$$

$$J = \{2\}$$

$$\left\{ \begin{array}{l} x_1 = +10 - 3 \cdot 3,5 \\ x_3 = 3,5 \\ x_2 = -5 + 2 \cdot 3,5 \end{array} \right.$$

$$I = \{1, 2, 3\}$$

$$J = \{1, 2, 3\}$$

$$\left\{ \begin{array}{l} x_1 = -0,5 \\ x_2 = 2 \\ x_3 = 3,5 \end{array} \right.$$

Example

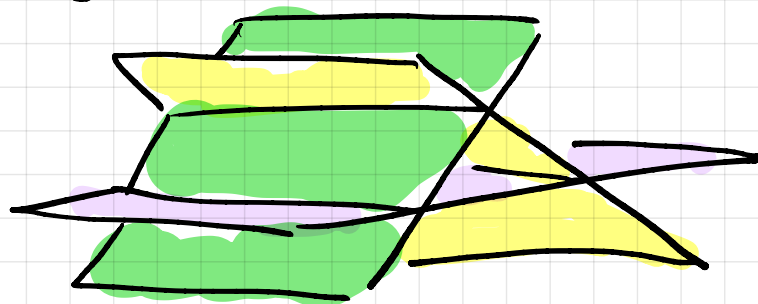
$$\begin{cases} x_1 + x_2 + x_3 = 5 & (1) \\ x_1 - x_2 + x_3 = 1 & (2) \\ 3x_1 + x_2 + 3x_3 = 0 & (3) \end{cases}$$



Gaussian Elimination

$$\begin{cases} x_1 + x_3 = 0 \\ x_2 = 0 \\ 0 = 1 \end{cases}$$

The system has no solutions.



Example To invert a map, given by $A \in F^{n \times n}$,
that sends $x \in F^n$ to $Ax = y \in F^n$
we need to solve the system $Ax = y$ in the
unknown y for an arbitrary choice of y .
That means, on the left-hand-side we have
unknowns $x_1, \dots, x_n$ and on the right-hand-side
we also have unknowns: $y_1, \dots, y_n$, and we
solve the system $Ax = y$ in the unknowns
 $x_1, \dots, x_n$.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \in \mathbb{Q}^{2 \times 2}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longmapsto \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\begin{cases} 1 \cdot x_1 + 2 \cdot x_2 = y_1 \\ 3 \cdot x_1 + 4 \cdot x_2 = y_2 \end{cases}$$

$$\begin{cases} 1 \cdot x_1 + 2 \cdot x_2 = 1 \cdot y_1 + 0 \cdot y_2 \\ 3 \cdot x_1 + 4 \cdot x_2 = 0 \cdot y_1 + 1 \cdot y_2 \end{cases}$$

One usually represents these equations with a matrix:

$$\begin{array}{cccc} & \begin{matrix} x_1 \\ \downarrow \end{matrix} & \begin{matrix} x_2 \\ \downarrow \end{matrix} & \begin{matrix} y_1 \\ \downarrow \end{matrix} & \begin{matrix} y_2 \\ \downarrow \end{matrix} \\ \text{1st equation} \rightarrow & 1 & 2 & 1 & 0 \\ \text{2nd equation} \rightarrow & 3 & 4 & 0 & 1 \end{array} \left[\begin{array}{cc|cc} \end{array} \right]$$

By transforming this system with the Gaussian-elimination, we arrive at an equivalent system

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right] \xrightarrow{-3 \times} \left\{ \begin{array}{l} x_1 + 2x_2 = y_1 \\ 3x_1 + 4x_2 = y_2 \end{array} \right.$$

$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right] \xrightarrow{+1 \times}$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & -2 & -3 & 1 \end{array} \right] \xrightarrow{\times (-\frac{1}{2})}$$

$$\left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array} \right]$$

$$\left\{ \begin{array}{l} x_1 = -2y_1 + y_2 \\ x_2 = \frac{3}{2}y_1 - \frac{1}{2}y_2 \end{array} \right.$$

So, the inverse of the linear map p
 $T: \mathbb{Q}^2 \rightarrow \mathbb{Q}^2$ given by

$T(x_1, x_2) = (x_1 + 2x_2, 3x_1 + 4x_2)$ is the map

$$T^{-1}: \mathbb{Q}^2 \rightarrow \mathbb{Q}^2 \text{ given by}$$

$$T^{-1}(y_1, y_2) = (-2y_1 + y_2, \frac{3}{2}y_1 - \frac{1}{2}y_2).$$

Example Consider the linear map

$$T: \mathbb{Q}^3 \rightarrow \mathbb{Q}^3 \text{ given by}$$

$$T(x) = A \cdot x \quad \text{where} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 5 & -1 & 8 \end{bmatrix}$$

$$T(x) = y, \text{ where}$$

$$\begin{cases} x_1 + x_2 + x_3 = y_1 \\ x_1 - x_2 + 2x_3 = y_2 \\ 5x_1 - x_2 + 8x_3 = y_3 \end{cases}$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & -1 & 2 & 0 & 1 & 0 \\ 5 & -1 & 8 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \curvearrowright \\ \downarrow \\ \downarrow \end{array}$$

Gaussian elimination

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -2 & 1 & -1 & 1 & 0 \\ 0 & -6 & 3 & -5 & 0 & 1 \end{array} \right] \begin{array}{l} \curvearrowright \\ \downarrow \\ \downarrow \end{array}$$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 2 & -1 & 0 \\ 0 & -2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & -3 & 1 \end{array} \right]$$

$$\begin{cases} x_1 + 3x_2 = 2y_1 - y_2 \\ -2x_2 + x_3 = -y_1 + y_2 \\ 0 = -2y_1 - 3y_2 + y_3 \end{cases}$$

There is no any $x \in \mathbb{Q}^3$ with

$$T(x) = y \text{ for } y = \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix}$$

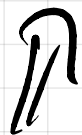
$\uparrow$
false

$$T(x) = y \iff$$

$$\begin{cases} x_1 + 3x_2 = 2 \cdot 1 - 1 \\ -2x_2 + x_3 = -1 + 1 \\ 0 = -2 \cdot 1 - 3 \cdot 1 + 5 \end{cases}$$



$$\begin{cases} x_1 + 3x_2 = 1 \\ -2x_2 + x_3 = 0 \\ 0 = 0 \end{cases}$$



$$\begin{cases} x_1 = 4 \\ x_2 = -1 \\ x_3 = -2 \end{cases}$$



$$x = \begin{bmatrix} 4 \\ -1 \\ -2 \end{bmatrix}$$

For $y = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, there is no any $x \in \mathbb{Q}^3$ such that $T(x) = y$.

Because

$$T(x) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (==)$$

$$\begin{cases} x_1 + 3x_2 = 2 \cdot 0 - 0 \\ -2x_2 + x_3 = -0 + 0 \\ 0 = -2 \cdot 0 - 3 \cdot 0 + 1 \end{cases}$$

$$0 = 1$$

The map is also not injective!

Let's consider all $x \in \mathbb{Q}^3$ with $T(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

By Gaussian elimination:

$$T(x) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$\begin{cases} x_1 + 3x_2 = 0 \\ -2x_2 + x_3 = 0 \\ 0 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = -3x_2 \\ x_3 = 2x_2 \\ x_2 \in \mathbb{Q} \end{cases}$$

So, T is not injective, because (for example)

$$T(0, 0, 0) = T(-3, 1, 2) = (0, 0, 0)$$

So, T is neither injective nor surjective.

Theorem 31 For a field F , every linear map $T: F^n \rightarrow F^n$ ($n \in \mathbb{N}$) is either bijective or neither injective nor surjective.

Proof is omitted, but see the above two examples.